Appendix A: Preprocessing Overview

We extracted reported p-values, sample sizes, and test information from the August 7, 2025 ClinicalTrials.gov snapshot. The final inclusion and exclusion criteria, analysis-level aggregation, and one-sided calculations are specified in Appendix C. Detailed preprocessing steps, code, and auxiliary clinical-trial plots are provided in the Online Supplement.

Appendix B: Supplementary Figures

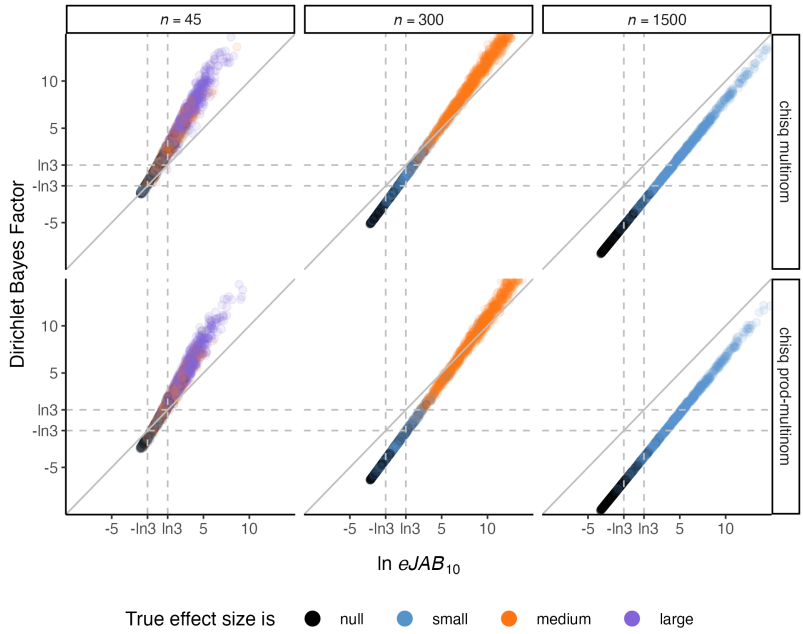


Figure S2: Comparison of the natural logarithmic $eJAB_{10}$ with the `contingencyTableBF` function from the '*BayesFactor*' `R` package at three total counts for chi-squared tests of independence under two sampling designs: multinomial (top) and product-multinomial (bottom). Colors represent the true effect size. The dashed lines indicate $eJAB_{10}$ of 1/3 and 3, representing moderate evidence against and for an effect, respectively.

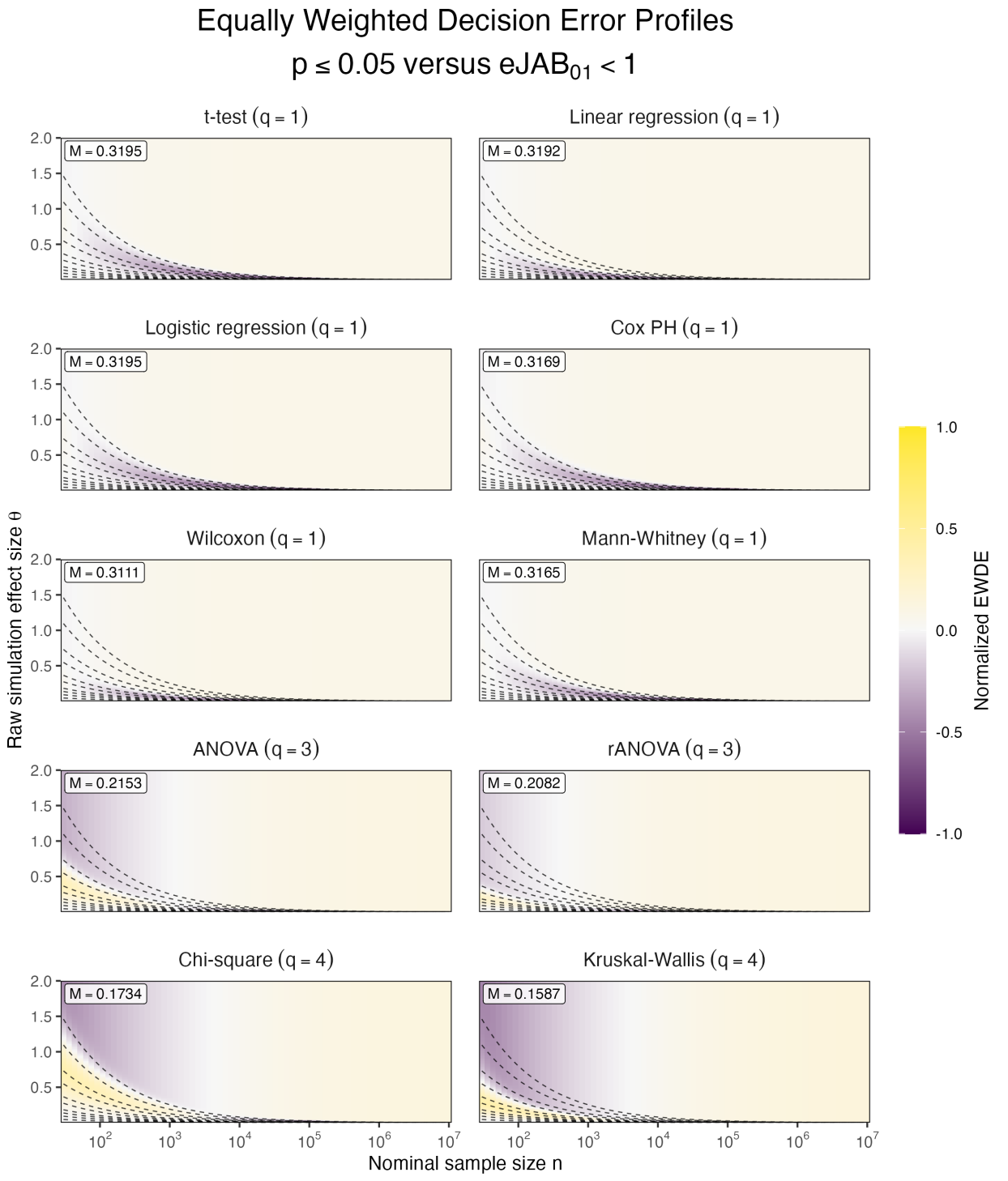


Figure S3: Equally Weighted Decision Error Profiles comparing the NHST significance rule $p \leq .05$ with $eJAB_{01} < 1$ across ten statistical tests. EWDE is the equally weighted average of the type I and type II error probabilities. Colors show $(\text{EWDE}_{\text{NHST}} - \text{EWDE}_{\text{eJAB}})/M$, where each panel reports its maximum absolute EWDE difference M over the displayed grid. Positive values (yellow) favor eJAB; negative values (purple) favor NHST. Dashed curves represent local alternatives of the form $\theta = c/\sqrt{n}$. The corresponding $p \leq .005$ comparison appears in Panel A of manuscript Figure 2.

Appendix C: Detailed ClinicalTrials.gov Inclusion and Exclusion Criteria

C.1 Scope, archived source, and units

The source was the ClinicalTrials.gov (CTG) JSON results export for completed studies with posted results, downloaded on August 7, 2025. The extraction emitted one row for every participant-denominator group attached to an outcome–analysis pair. A deterministic MD5 hash of the CTG identifier, the complete outcome object, and the complete analysis object (with sorted JSON keys) defined `analysisId`. Thus, `analysisId`, rather than an individual denominator row or a CTG study, was the unit of statistical analysis. Rows sharing an `analysisId` were group-denominator components of the same test and were collapsed during sample-size construction. Statistical analyses were nested within CTG studies identified by `nctId`.

The manuscript analysis estimates the frequency of disagreement between NHST significance and eJAB evidence among eligible primary- and secondary-outcome analyses from exact Early Phase I, exact Phase I, and exact Phase II interventional trials. Both one-sided and two-sided/unrestricted tests are included, using $p_1 \leq .025$ and $p \leq .05$, respectively. The three phase classifications are retained as separate groups. These criteria define the screened set used throughout the manuscript and supplement.

C.2 Inclusion criteria

An analysis was eligible only when all of the following criteria were met:

1. The record occurred in the archived CTG results extraction and had a deterministic `analysisId` and `nctId`.
2. CTG classified the study type as `INTERVENTIONAL`.
3. CTG classified the study as *exactly* Early Phase I, *exactly* Phase I, or *exactly* Phase II. Exact Early Phase I required `EARLY_PHASE1`=1 and `PHASE1`=`PHASE2`=`PHASE3`=`PHASE4`=0. Exact Phase I and exact Phase II were defined analogously using their respective indicator.
4. CTG labeled the outcome as `PRIMARY` or `SECONDARY`.
5. The reported p-value contained a numeric value strictly between 0 and 1 and was not right-censored (for example, $p > .05$).
6. The free-text statistical method mapped deterministically to one of the eJAB-computable classes in Table S2.
7. All participant denominators required for the mapped test were finite and positive; the derived sample size satisfied $n > 1$; and the test dimension satisfied $q \geq 1$.
8. The resulting direction-appropriate eJAB was finite and positive.

Statistical significance was *not* an eligibility criterion. Both significant and nonsignificant eligible results were retained for the descriptive analysis.

C.3 Exclusion criteria

The following were excluded, in the sequential order shown in Table S1:

- Four prespecified malformed analysis identifiers previously found to contain denominator errors.
- Observational and expanded-access records, and any record not labeled interventional.
- Phase I/Phase II, Phase II/Phase III, Phase III, Phase IV, not-applicable, or missing phase classifications. Combined phases were not recoded into any of the three early-phase groups.
- Outcomes other than primary or secondary, baseline measures, participant-flow analyses, and adverse-event analyses.

- Missing, blank, nonnumeric, zero, one, or otherwise invalid p-values. Right-censored values were excluded because their numerical bound does not identify the value required for eJAB. Left-censored values were handled as described in Section C.5.
- Methods outside the prespecified mappings, including ANCOVA, mixed-effects or repeated-measures models, Fisher exact tests, chi-squared tests without full table dimensions, Cox or log-rank analyses without the number of observed events, Bayesian analyses, and other free-text methods that could not be mapped without assumptions not supported by the CTG record.
- Analyses with missing, zero, or ambiguous participant denominators; one-sample or signed-rank analyses represented by more than one denominator row; omnibus analyses represented by fewer than two groups; $n \leq 1$; or $q < 1$.
- A nonfinite or nonpositive eJAB value after all preceding steps.

Table S1: Sequential inclusion and exclusion flow. Counts are analysis identifiers except in the last column. Each exclusion count is conditional on all previous steps.

Screening stage	Excluded	Retained	Studies
Unique analyses in the archived extraction	0	279,927	22,454
Remove four malformed analysis identifiers	4	279,923	22,453
Restrict to interventional studies	11,893	268,030	21,500
Restrict to exact Early Phase I, Phase I, or Phase II	175,191	92,839	6,302
Restrict to primary and secondary outcomes	1,806	91,033	6,277
Require an admissible p-value and exclude right-censoring	26,665	64,368	4,461
Require a prespecified eJAB method mapping	47,214	17,154	1,877
Require valid denominators, row structure, n, and q	591	16,563	1,790

C.4 Method mapping, dimension, and sample size

Free-text method labels were normalized using case-insensitive regular expressions. More specific patterns were evaluated before broader patterns (for example, conditional logistic regression before logistic regression, and one-sample or paired t-tests before other t-tests). Table S2 gives the complete set used for the early-phase reanalysis.

Table S2: Eligible statistical method classes and derived quantities. Here G is the number of attached denominator groups.

Mapped class	Sample-size rule	Dimension q
One-sample or paired t-test	One attached denominator; n is that value	1
Two-sample t-test	Sum attached group denominators	1
Linear regression	Sum attached denominators	1
Logistic regression	Sum attached denominators	1
Conditional logistic regression	Maximum attached denominator	1
One-way ANOVA	Sum attached group denominators; require $G \geq 2$	$G - 1$
Wilcoxon signed-rank test	One attached denominator; n is that value	1
Mann–Whitney test	Sum attached group denominators	1
Kruskal–Wallis test	Sum attached group denominators; require $G \geq 2$	$G - 1$

Participant counts containing thousands separators were parsed as numbers before aggregation. A missing or nonpositive attached denominator made the full analysis ineligible; missing values were not silently removed from a sum. The archived extraction's repeated `analysisId` rows encode attached denominator groups and were therefore aggregated, not counted as duplicate hypothesis tests. The hash construction produced one final record per outcome–analysis pair; no additional duplicate analysis identifiers remained after aggregation.

C.5 Reported inequalities, test direction, and Wagenmakers's one-sided Bayes factor

The numeric component of a reported p-value was extracted reproducibly. For a left-censored value such as $p < .05$, the bound .05 was used in the eJAB calculation. This is conservative for evidence against the null because the unknown underlying value is smaller. The reported inequality itself was retained: a report of $p < .05$ met the $p \leq .05$ decision rule even though the stored numerical bound equaled .05. Numerical reports were compared as displayed, so a reported $p = .05$ was also included (and likewise $p_1 = .025$ for a one-sided test). A rounded value at the threshold does not establish that the underlying full-precision value satisfies the rule; these classifications describe the reported values. Values reported with ">" were excluded because a lower bound does not identify the tail probability needed for eJAB. The final eligible set contained 2,714 left-censored values (30 in Early Phase I, 429 in Phase I, and 2,255 in Phase II).

Direction was determined from the reported `statisticalMethod`. The patterns `one-sided`, `1-sided`, `one-tailed`, and `1-tailed` identified a one-sided test. A directional correction was used only for scalar tests ($q = 1$). An explicit two-sided label or the absence of an explicit one-sided method label was classified as two-sided/unrestricted. Mentions of one-sided α found only in sample-size or power prose were not allowed to reclassify the reported test. This rule identified 427 one-sided analyses and 16,136 two-sided/unrestricted analyses.

For a one-sided reported value p_1, the corresponding two-sided value was first reconstructed as

$$p = 2\min\{p_1, 1 - p_1\}.$$

For two-sided/unrestricted analyses, eJAB was computed as

$$eJAB_{01} = \sqrt{n}\exp\left\{-\frac{1}{2}\left(1 - n^{-1/q}\right) Q_{\chi^2_q}(1-p)\right\}.$$

For one-sided scalar analyses, we used the null-centered definition in the manuscript, combining the Savage–Dickey approximation with the directional correction of Wagenmakers (2022, pp. 20–22):

$$eJAB_{0+} = \frac{\sqrt{n}}{2(1-p_1)}\exp\left\{-\frac{1}{2}Q_{\chi^2_1}(1-p)\right\}.$$

This null-centered construction omits the finite-sample factor $(n-1)/n$ from the exponent. The directional correction uses prior mass $1/2$ and approximate posterior mass $1 - p_1$ in the predicted direction for moderate samples and a relatively vague local prior. Thus the procedure does not merely substitute a one-sided p-value into a chi-squared formula.

Statistical significance was defined as $p_1 \leq .025$ for one-sided analyses and $p \leq .05$ for two-sided/unrestricted analyses. Plots use two-sided p-values, with $p = 2\min\{p_1, 1 - p_1\}$ for one-sided tests. At each sweep value α, a *Bayes/NHST contradiction* requires $p_1 \leq \alpha/2$ for a one-sided test or $p \leq \alpha$ otherwise, together with the applicable eJAB exceeding 1. The directional rule excludes opposite-tail results even when their folded p-value is small. This descriptive term does not assert that the result is a type I error.

C.6 Final analytic set and reproducibility

The final set comprised 16,563 primary- and secondary-outcome analyses from 1,790 trials: 270 analyses from 47 exact Early Phase I trials, 2,098 analyses from 280 exact Phase I trials, and 14,195 analyses from 1,463 exact Phase II trials. Combining one-sided tests with $p_1 \leq .025$ and two-sided/unrestricted tests with $p \leq .05$, 94 Early Phase I, 927 Phase I, and 5,453 Phase II analyses were NHST significant. Six Early Phase I results (6.38% of all significant results in that phase), 28 Phase I results (3.02%), and 459 Phase II results (8.42%) were Bayes/NHST contradictions, with the applicable eJAB > 1. The significant denominators include 4, 27, and 104 one-sided results, respectively; none has one-sided eJAB > 1.

The complete executable pipeline is `analysis/early_phase/early_phase_analysis.R` in the GitHub repository. It writes the sequential flow table, the eligible analysis-level data set, phase and test summaries, and the clustered mixed-model sensitivity result. The four manual exclusions, method regular expressions, direction rules, significance thresholds, denominator rules, and eJAB calculations are encoded directly in that script. The supplementary JLP catalogue is generated from this eligible set by `analysis/early_phase/jlp_catalog.R`; its machine-readable output retains the complete analysis identifiers and archived source descriptions.

The July 2026 protocol was archived after the August 2025 CTG snapshot had already been downloaded and inspected. Accordingly, this is described as a protocol-aligned retrospective reanalysis, not as a prospective confirmatory study or an independent replication. For phase g, let N_g count all eligible analysis records, S_g those meeting the NHST significance rules in Section C.5, and $R_g(K)$ those significant records with applicable eJAB $> K$. Figure S7A plots the overall proportion $R_g(K)/N_g$ as the BF threshold K varies, marking contradictions at $K = 1$ and the stronger Jeffreys–Lindley screen at $K = 3$. The corresponding observed proportion among significant records is $R_g(K)/S_g$. These summaries weight analysis records equally within phase, rather than averaging trial-specific proportions or estimating the fraction of trials containing a flag.

For $K = 3$, the conditional proportions are 0/94 (0%), 2/927 (0.22%), and 18/5,453 (0.33%); the corresponding overall proportions are 0/270 (0%), 2/2,098 (0.095%), and 18/14,195 (0.127%). The 20 screened records include the reporting and hypothesis qualifications described in Section C.7. Both Phase I records are bound-only reports; Phase II contains 11 numerical reports and seven bound-only reports. Posterior summaries for the proportion with eJAB > 1 among NHST-significant results are provided in Appendix F. Figure S6 shows the complementary NHST significance-level sweep at the fixed BF threshold 1. Figure S8 provides an enlarged scatterplot using two-sided p-values over $[0, 1]$.

C.7 Catalogue of screened Jeffreys–Lindley cases

We screened the eligible set for reported NHST significance ($p_1 \leq .025$ for one-sided tests and $p \leq .05$ otherwise), together with the applicable eJAB > 3. This stronger disagreement threshold is a descriptive criterion for identifying finite-sample examples compatible with the Jeffreys–Lindley phenomenon under the stated eJAB approximation; it does not establish that a scientific effect is absent. The screen returns 20 analysis records from nine trials: two in Phase I and 18 in Phase II, with none in Early Phase I. All 20 were mapped to two-sided/unrestricted scalar tests ($q = 1$) by the method-label rule.

The catalogue distinguishes numerical reports from bounds. Eleven records, all in Phase II and drawn from five trials, have numerical reported p-values and yield point-calculated eJAB values above 3. The other nine report only $p < .05$: evaluating eJAB at .05 gives an *upper bound* on the null evidence, so eJAB > 3 cannot be established for those records. Four of these nine also have equivalence or bioequivalence context in the archived analysis descriptions; the generic method-label mapping does not justify interpreting their p-values as ordinary tests of the point null used by eJAB. They are retained in the screening catalogue for transparency and are not presented as confirmed JLP examples. Numerical reports may themselves be rounded, and the classification refers to the displayed values, not unavailable full-precision values. The recorded method labels and available archived descriptions also do not independently verify every hypothesis specification or complete-case sample size.

Table S3 gives each reported finding and comparison, its method, participant-based sample size, reported p-value, eJAB value or bound, and qualification. Separate outcome–analysis records from one trial are retained as separate rows and are not independent trial replications. Linked registry identifiers allow inspection of the trial; the descriptions and calculations here use the archived August 7, 2025 extract, which may differ from the current registry record.

Table S3: Early-phase records with reported NHST significance and computed $eJAB_{01} > 3$. The screening set includes literal p-value bounds; flagged rows require the qualifications below.

Trial / phase	Outcome or comparison	Method / n	Reported p	$eJAB_{01}$	Flag
J01 NCT00071513 Phase I	Change in Short Moods and Feelings Questionnaire (SMFQ)	Two-sample t-test $n = 480$	< 0.05	≤ 3.23	B
J02 NCT02260492 Phase I	FEV1 Trough	ANOVA $n = 801$	< 0.05	≤ 4.16	B,E
J03 NCT00405821 Phase II	HIV-1 Viral Load Difference Between Arms	Two-sample t-test $n = 440$	0.05	3.09	N
J04 NCT00565812 Phase II	Number of Participants With Decrease in Total Analgesic Medication Use	Logistic regression $n = 971$	0.032	3.13	N
J05 NCT00565812 Phase II	Number of Participants With Joint Space Narrowing Progression	Logistic regression $n = 848$	0.047	4.06	N
J06 NCT01225822 Phase II	Major bleeding events: BIBR 1048 50 mg twice daily versus enoxaparin 40 mg daily	Logistic regression $n = 781$	0.0472	3.91	N
J07 NCT01225822 Phase II	Proximal DVT: BIBR 1048 225 mg versus 50 mg, both twice daily	Logistic regression $n = 753$	0.036	3.05	N

Trial / phase	Outcome or comparison	Method / n	Reported p	$eJAB_{01}$	Flag
J08 NCT01225822 Phase II	Proximal DVT, pulmonary embolism and VTE-related mortality: BIBR 1048 225 mg versus 50 mg, both twice daily	Logistic regression $n = 753$	0.0373	3.15	N
J09 NCT01225822 Phase II	VTE events and all-cause mortality: BIBR 1048 150 mg twice daily versus enoxaparin 40 mg daily	Logistic regression $n = 757$	0.0401	3.36	N
J10 NCT01225822 Phase II	VTE events: BIBR 1048 150 mg twice daily versus enoxaparin 40 mg daily	Logistic regression $n = 757$	0.0401	3.36	N
J11 NCT01434680 Phase II	Meningococcal C hSBA geometric mean titers at day 28: MenC-CRM ROS versus EMV	ANOVA $n = 540$	< 0.05	≤ 3.42	B,E
J12 NCT01434680 Phase II	Meningococcal C hSBA geometric mean titers at day 28: MenC-CRM LIQ versus EMV	ANOVA $n = 564$	< 0.05	≤ 3.50	B,E
J13 NCT01434680 Phase II	Meningococcal C hSBA geometric mean titers at day 28: MenC-CRM liquid versus ROS	ANOVA $n = 542$	< 0.05	≤ 3.43	B,E
J14 NCT02303704 Phase II	Pharamacological Inotropic Support (Dobutamine)	Two-sample t-test $n = 434$	< 0.05	≤ 3.07	B
J15 NCT02303704 Phase II	Pharmacologic Inotropic Support (Adrenaline)	Two-sample t-test $n = 434$	< 0.05	≤ 3.07	B
J16 NCT02303704 Phase II	Pharmacological Inotropic Support (Nor-adrenaline)	Two-sample t-test $n = 434$	< 0.05	≤ 3.07	B
J17 NCT02303704 Phase II	Post-op CK-MB Levels	Two-sample t-test $n = 434$	< 0.05	≤ 3.07	B
J18 NCT04376684 Phase II	Part 1: Percentage of Participants Alive and Free of Respiratory Failure at Day 28	Logistic regression $n = 782$	0.0456	3.80	N
J19 NCT05013008 Phase II	Plasma biomarker change: CCN family member 4 (O95388)	Two-sample t-test $n = 921$	$= 0.044$	4.00	N
J20 NCT05013008 Phase II	Plasma biomarker change: TGF beta-1 proprotein (P01137)	Two-sample t-test $n = 921$	$= 0.044$	4.00	N

N: numeric p report; B: left-censored p report. For B rows, the displayed eJAB is evaluated at the reported upper bound and is itself an upper bound (rounded upward for display), so $eJAB_{01} > 3$ is not established. E: archived descriptions indicate equivalence or bioequivalence hypotheses; an ordinary point-null interpretation is not justified without further review. NCT02260492 refers to a Day 1 analysis that explicitly describes two one-sided bioequivalence tests. All listed analyses were classified as two-sided/unrestricted in the screening pipeline and have $q = 1$; significance includes $p = .05$. Participant counts are sums of the distinct outcome-level groups associated with each analysis; they do not independently verify the analysis-specific complete-case sample size. Multiple outcomes or biomarkers within one trial are separate analyses, not independent trials. Numeric reports remain subject to rounding and the approximation's assumptions. Row identifiers J01–J20 map to the complete archived analysis identifiers in `jlp_cases.csv`.

Appendix D: Proof of Model-Selection Consistency

Lemma 1. *For $x \to 1^-$,*

$$0 < f(x) = -2\ln(1-x) - 2\ln(-\ln(1-x)) < Q_{\chi^2_q}(x).$$

The proof is provided in the GitHub repository (see files within 'Theorem').

PROOF OF THEOREM 2.3. Assume $\mathcal{H}_0$ is true. Let $\delta \in (0,1)$ and $K > 0$ be arbitrary. Note that **R1** and $W_n = Q_{\chi^2_q}(1-p) \xrightarrow{D} \chi^2_q$ by the continuous mapping theorem.
For $n \geq N^* = \lceil K^2 \exp\{Q_{\chi^2_q}(1-\delta)\} \rceil$, we have

$$\begin{aligned}
&\mathbb{P}(eJAB_{01} > K) = \mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\frac{n^{1/q}-1}{n^{1/q}}\,W_n\right\} > K\right)\\
>&\mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\,W_n\right\} > K\right)\\
=&\mathbb{P}\left(W_n < -2\ln\frac{K}{\sqrt{n}}\right)\\
\geq&\mathbb{P}\left(W_n < -2\ln\frac{K}{\sqrt{N^*}}\right)\\
\geq&\mathbb{P}\left(W_n < -2\ln\frac{K}{\sqrt{K^2\exp\{Q_{\chi^2_q}(1-\delta)\}}}\right)\\
=&\mathbb{P}\left(W_n < Q_{\chi^2_q}(1-\delta)\right) \to 1-\delta \text{ as } n \to \infty.
\end{aligned}$$

Thus, $eJAB_{01} \xrightarrow{P} \infty$ as $n \to \infty$ under $\mathcal{H}_0$.

Assume $\mathcal{H}_1$ is true. Let $\delta \in (0,1)$ and $\epsilon > 0$ be arbitrary. Lemma 1 gives $0 < f(1-p) < Q_{\chi^2_q}(1-p)$ for $p \to 0$.

$$\begin{aligned}
&\mathbb{P}(eJAB_{01} < \epsilon)\\
=&\mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\frac{n^{1/q}-1}{n^{1/q}}\,Q_{\chi^2_q}(1-p)\right\} < \epsilon\right)\\
>&\mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\frac{n^{1/q}-1}{n^{1/q}} f(1-p)\right\} < \epsilon\right)\\
=&\mathbb{P}\left(\sqrt{n}\ \left(p\cdot(-\ln p)\right)^{\frac{n^{1/q}-1}{n^{1/q}}} < \epsilon\right) > 1-\delta.
\end{aligned}$$

The final inequality is true for n sufficiently large by **R2** and the continuous mapping theorem with $1 - \frac{1}{n^{1/q}} \to 1$.

Thus, $eJAB_{01} \xrightarrow{P} 0$ as $n \to \infty$ under $\mathcal{H}_1$. Together, *eJAB* is model-selection consistent.

□

Appendix E: Realized Values of D_n in Four Replication Projects

Considering **R2** of Theorem 2.3 from a practical standpoint, we examine the realized values of $D_n = -\sqrt{n} \cdot p \ln p$ in four large-scale replication projects. Each project selected a collection of previously published findings and attempted to replicate them under a preregistered protocol, so that each original result was paired with an independently conducted replication of the same effect.

- **EERP**, the Experimental Economics Replication Project (Camerer et al., 2016, *Science*), which attempted to replicate laboratory experiments published in the *American Economic Review* and the *Quarterly Journal of Economics* between 2011 and 2014;
- **RPP**, the Reproducibility Project: Psychology (Open Science Collaboration, 2015, *Science*), which attempted to replicate a collection of findings drawn from three high-impact psychology journals;
- **RPCB**, the Reproducibility Project: Cancer Biology (Errington et al., 2014, *eLife*), which considered experiments reported in high-impact preclinical cancer biology papers; we use the project's final analysis (Errington et al., 2021, *eLife*); and
- **SCORE**, Systematizing Confidence in Open Research and Evidence (Tyner et al., 2025, *Nature*), which assembled and considered replication of a large corpus of published claims from the social and behavioral sciences.

For each result in the four projects, we extracted the original p-value and sample size n, and the p-value and sample size of each of its replication attempts. Each result has at least one and possibly several replication attempts. A result is included in the $\mathcal{H}_1$ subset when its original p-value satisfies $p \leq 0.005$ and every one of its replication attempts also reached $p \leq 0.005$. Given that the effect was found at this level in the original study and in every replication, we make the assumption that $\mathcal{H}_1$ is true for these results. This gives realized values of D_n under $\mathcal{H}_1$: one from each original study and one from each replication attempt. Table S4 reports the counts at each step.

Two kinds of recorded p-value were treated separately. Where a p-value was recorded as 0, we computed $\ln p$ from the reported test statistic and, where the test has them, its degrees of freedom. Of the plotted values, 51 were obtained this way, including the value at the largest sample size; 12 values (six SCORE original results and six SCORE replication attempts) could not be computed and were dropped. Where a p-value was reported as a bound, such as $p < .001$, rather than as a value, we removed it. Such bounds occur only in SCORE. SCORE labels its original p-values; we removed the 37 it labels as less-than or threshold values, and the one with no label. SCORE does not label replication p-values, so we treated a replication p-value recorded as an exact power of ten, such as .001, as censored unless we were able to recompute the p-value directly from the reported test statistic. This removed two replication values where the p-value is left censored and cannot be recalculated from the reported test statistic.

The columns of Table S4 are as follows.

(1) *Released*: the number of results in each project's released data. A result is a study for EERP (18) and RPP (167, of which 100 were replicated), an effect for RPCB (158 effects from 23 papers, given in 188 rows), and a claim for SCORE (331 claims with at least one replication report).
(2) *With p and n*: the results in column 1 for which both the original p-value and the original sample size are recorded.
(3) *Original $p \leq 0.005$*: the results in column 2 whose original p-value satisfies $p \leq 0.005$.
(4) *$\mathcal{H}_1$ subset*: the results in column 3 for which every replication attempt reached $p \leq 0.005$. A result with no recorded replication p-value is not included; this excludes 20 RPP results and one RPCB result. We also excluded RPP study 69. The RPP data record $p = 10^{-6}$ for both its original and its replication, with no test statistic from which either value could be checked, whereas the reported p-values in the same data are $p < .05$ for the original and $p = .05$ for the replication. For SCORE, a replication attempt is one replication report.
(5) *Plotted values*: the values of D_n shown in Figure S4, one for each original result and one for each

replication attempt in the $\mathcal{H}_1$ subset, less the p-values reported as bounds and the zero p-values whose $\ln p$ could not be computed. For RPCB and SCORE the replication column exceeds the $\mathcal{H}_1$ subset because a result can have more than one replication attempt (up to three in RPCB and four in SCORE), each contributing its own value.

Project	Field	Results				(5) Plotted values	
		(1) released	(2) with p, n	(3) $p \leq 0.005$	(4) $\mathcal{H}_1$ subset	original	replication
EERP	Experimental economics	18	18	7	5	5	5
RPP	Psychology	167	143	67	16	16	16
RPCB	Preclinical cancer biology	158	115	61	15	15	21
SCORE	Social and behavioral sciences	331	311	184	102	58	107
	Total	674	587	319	138	94	149

Table S4: Counts at each step of the selection, by project. The columns are described in the notes above the table.

Figure S4 plots the realized values of D_n against the sample size n, which spans roughly seven orders of magnitude. The most important of the realized values is the one at the largest sample size, which is circled in the figure. Based on this, the condition **R2** seems feasible. The data, code and counts for Table S4 and Figure S4 are in the GitHub repository (see 'Supplementary Information/R2Empirical').

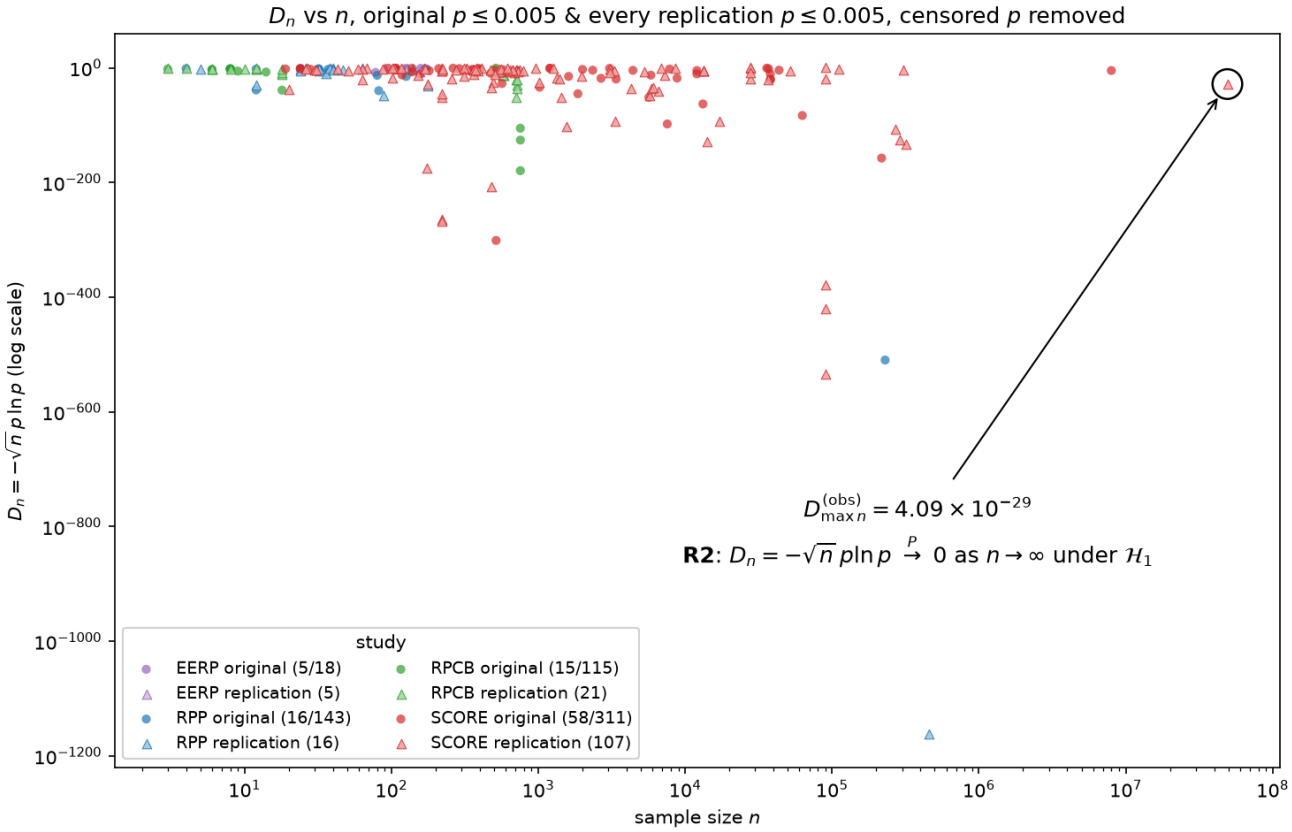


Figure S4: Realized values of $D_n = -\sqrt{n} \cdot p \ln p$, on a logarithmic scale, plotted against sample size n for the $\mathcal{H}_1$ subsets of the four replication projects (original $p \leq 0.005$ and every replication attempt $p \leq 0.005$). Circles denote original results and triangles replication attempts, with symbols colored by project. Legend entries give the number of plotted values; for original results this is shown relative to column 2 of Table S4. p-values reported as bounds are excluded. The circled value, $D_{\max n}^{(\text{obs})} = 4.09 \times 10^{-29}$, is the observed D_n at the largest sample size.

Appendix F: Posterior sensitivity

Within each phase g, let $\boldsymbol{y}_i = (y_{i1}, \ldots, y_{i4})$ be trial i's counts in the four categories formed by NHST significance and eJAB > 1: non-significant/eJAB ≤ 1, non-significant/eJAB > 1, significant/eJAB ≤ 1, and significant/eJAB > 1. With independent Dirichlet(a, a, a, a) priors, the posterior category probabilities satisfy

$$\boldsymbol{\pi}_i \mid \boldsymbol{y}_i \sim \text{Dirichlet}(y_{i1} + a, \ldots, y_{i4} + a).$$

Writing $m_i = \sum_j y_{ij}$ for each trial's eligible-analysis count, we summarize the conditional proportion

$$\rho_g = \frac{\sum_{i \in g} m_i \pi_{i4}}{\sum_{i \in g} m_i (\pi_{i3} + \pi_{i4})}.$$

We use 50,000 posterior draws for each $a \in \{10^{-2}, 10^{-4}, 10^{-6}\}$. Figure S5 shows kernel density estimates, and Table S5 reports posterior means and 95% equal-tailed credible intervals. The calculation is implemented in `analysis/early_phase/posterior_prior_sensitivity.R` in the GitHub repository.

Table S5: Posterior means and 95% equal-tailed credible intervals for the fixed-cohort contradiction proportion. All entries are percentages. Each prior–phase combination uses 50,000 draws.

Phase	Observed	$a = 10^{-2}$	$a = 10^{-4}$	$a = 10^{-6}$
Early Phase I	6.383	6.753 [3.894, 10.305]	6.382 [3.690, 9.726]	6.384 [3.734, 9.697]
Phase I	3.020	3.282 [2.717, 3.919]	3.024 [2.528, 3.585]	3.021 [2.529, 3.581]
Phase II	8.417	8.629 [8.012, 9.274]	8.417 [7.806, 9.041]	8.419 [7.812, 9.057]

Posterior mean Monte Carlo standard errors are at most 0.0074 percentage points; batch-based interval-endpoint Monte Carlo standard errors are at most 0.0274 percentage points. For 10^{-4} versus 10^{-6}, posterior mean changes are below 0.004 percentage points in every phase. The 10^{-2} means exceed the 10^{-6} means by 0.369, 0.261, and 0.211 percentage points, respectively.

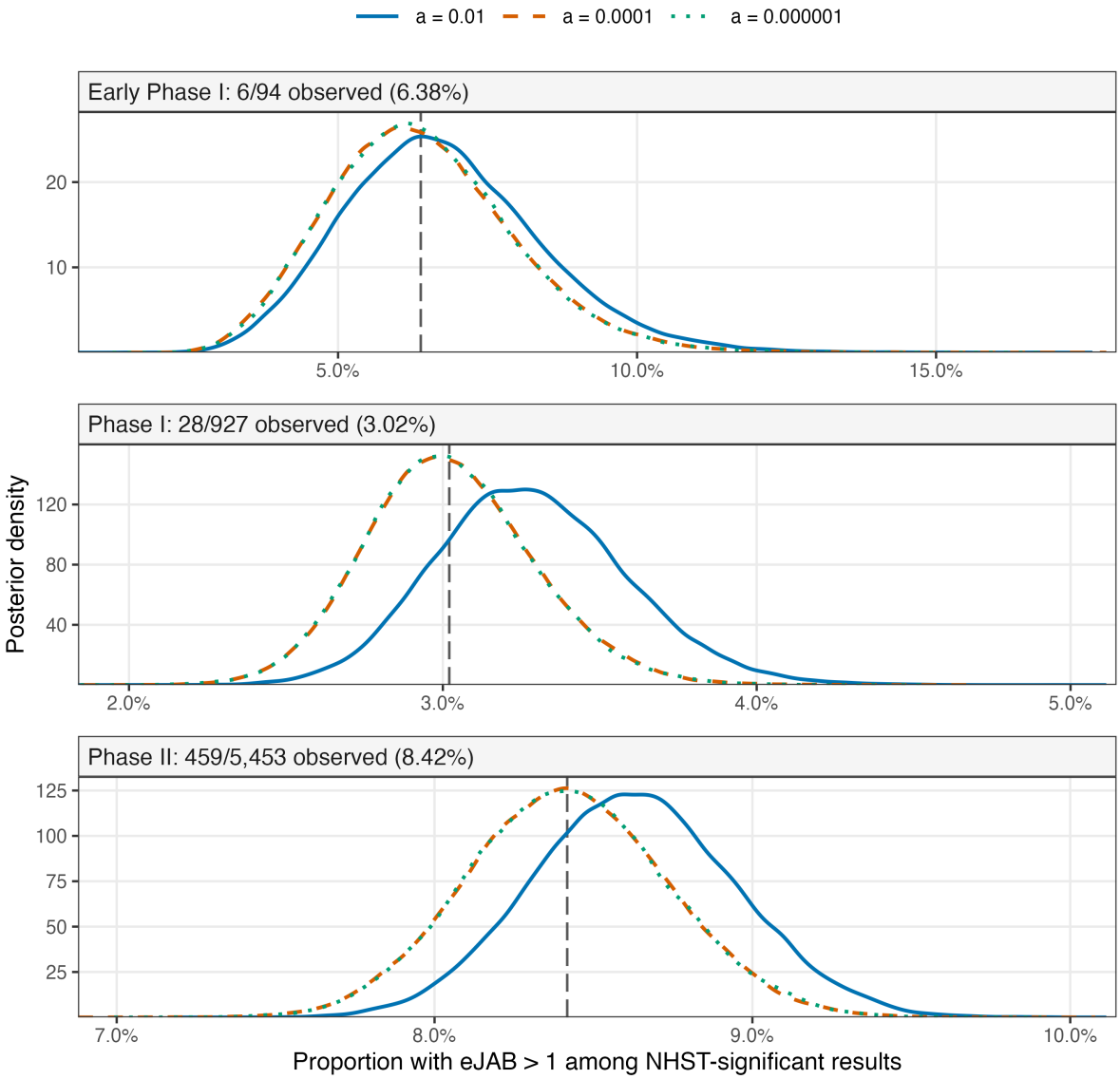


Figure S5: Posterior sensitivity for the proportion of NHST-significant records with eJAB > 1. Each phase panel overlays kernel density estimates from 50,000 posterior draws under independent per-trial Dirichlet(a, a, a, a) priors, with eligible-analysis-count weights. Grey dashed lines mark observed screening proportions; panel axes differ. The eligible set, NHST rule, and eJAB threshold are held fixed across priors. Small symmetric concentration parameters favor boundary configurations; the curves assess sensitivity within this prior family.

Appendix G: NHST significance-level profiles

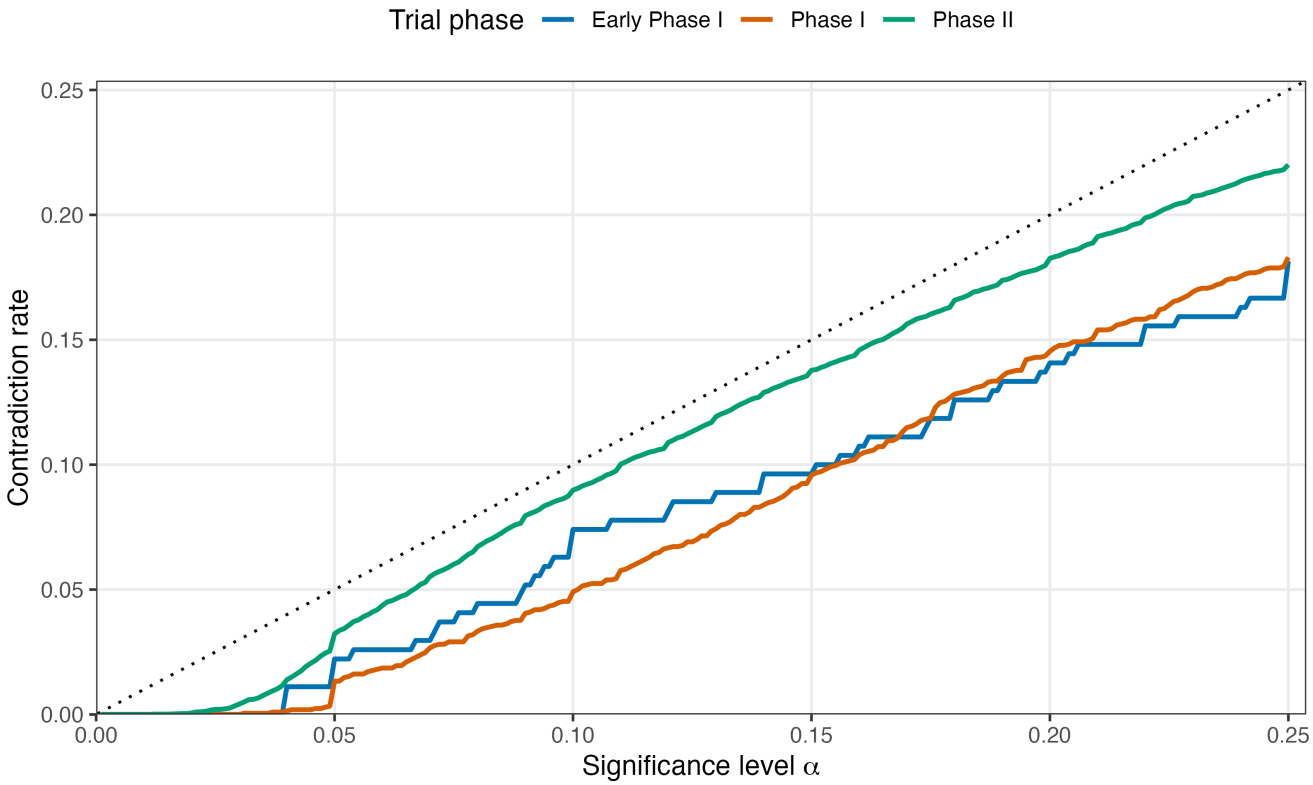


Figure S6: Overall Bayes/NHST contradiction proportions across nominal NHST significance levels, with separate curves for Early Phase I, Phase I, and Phase II. For phase g, the numerator $C_g(\alpha)$ counts eligible analyses with $p_1 \leq \alpha/2$ for one-sided tests or $p \leq \alpha$ for two-sided tests, together with applicable eJAB > 1, and the denominator N_g is all eligible analyses in that phase. The dotted line marks equality with α. Reported values and upper bounds at .05 produce a jump; the curves describe screening frequency, not type I error rates. At $\alpha = .05$, these are the same overall proportions shown at $K = 1$ in Figure S7A. Figure S5 instead summarizes the conditional proportion among NHST-significant results.

Appendix H: Clinical-trial profiles

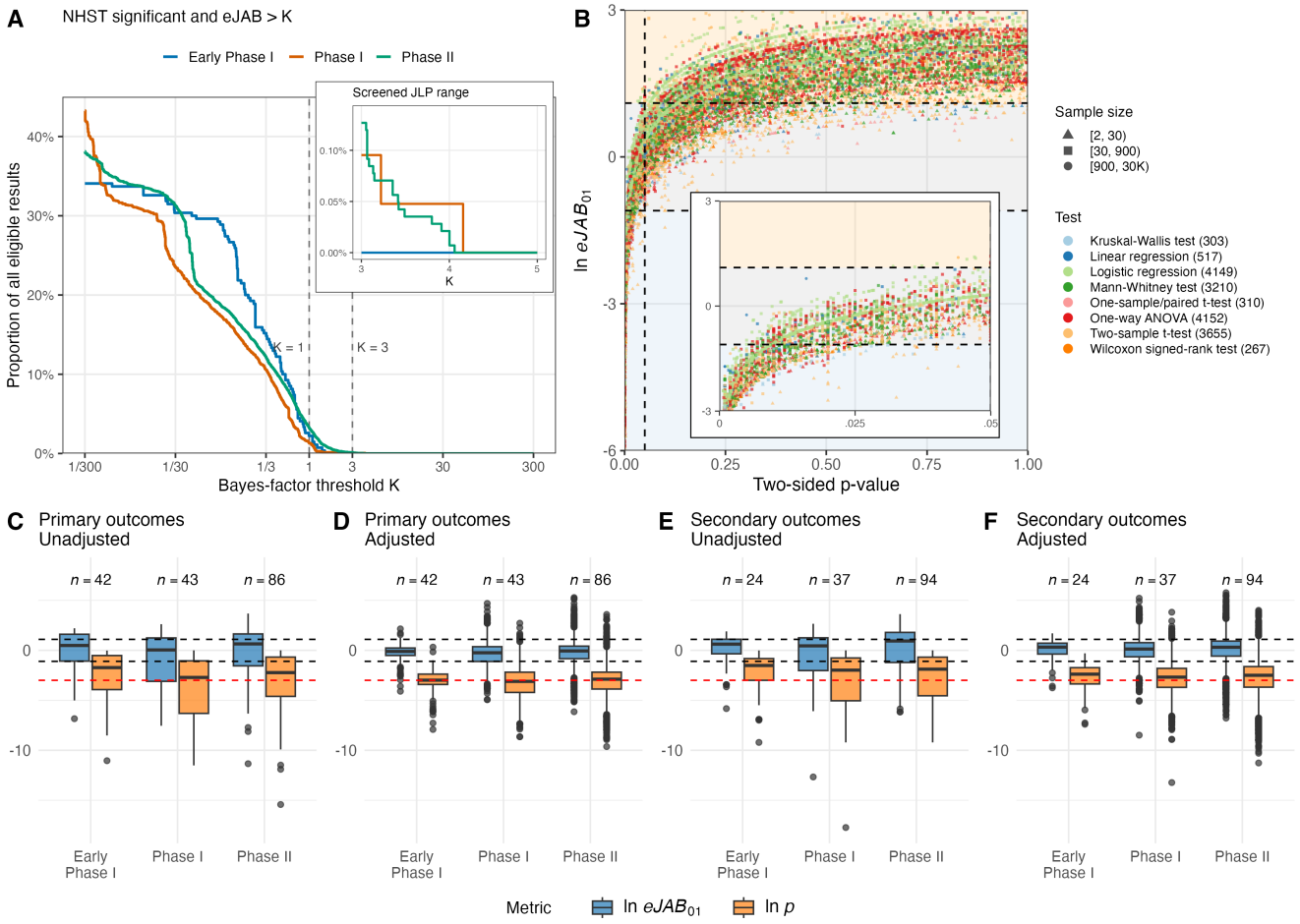


Figure S7: Clinical-trial summaries by Early Phase I, Phase I, and Phase II, combining one-sided and two-sided/unrestricted tests. NHST significance requires $p_1 \leq .025$ for one-sided tests and $p \leq .05$ otherwise. (A) Proportion of all eligible analyses meeting these rules with applicable eJAB $> K$, versus the BF threshold K. The marked thresholds are 1 (contradictions) and 3 (screened Jeffreys–Lindley cases); the inset enlarges $K \in [3, 5]$. (B) $\ln eJAB_{01}$ versus the two-sided p-value, using $p = 2\min\{p_1, 1 - p_1\}$ for one-sided tests, with statistical-test colors and sample-size shapes. Dashed lines mark .05 and eJAB $= 1/3, 3$; the inset shows $p \leq .05$ and $-3 \leq \ln eJAB_{01} \leq 3$. (C, E) Unadjusted distributions of $\ln eJAB_{01}$ (blue) and $\ln p$ (orange) for primary and secondary outcomes, using the same two-sided p-values. (D, F) Corresponding mixed-model-adjusted distributions. Panels C–F share a y-axis scale and show average analysis sample sizes; black dashed lines mark $\pm \ln 3$ and red dashed lines mark $\ln 0.05$.

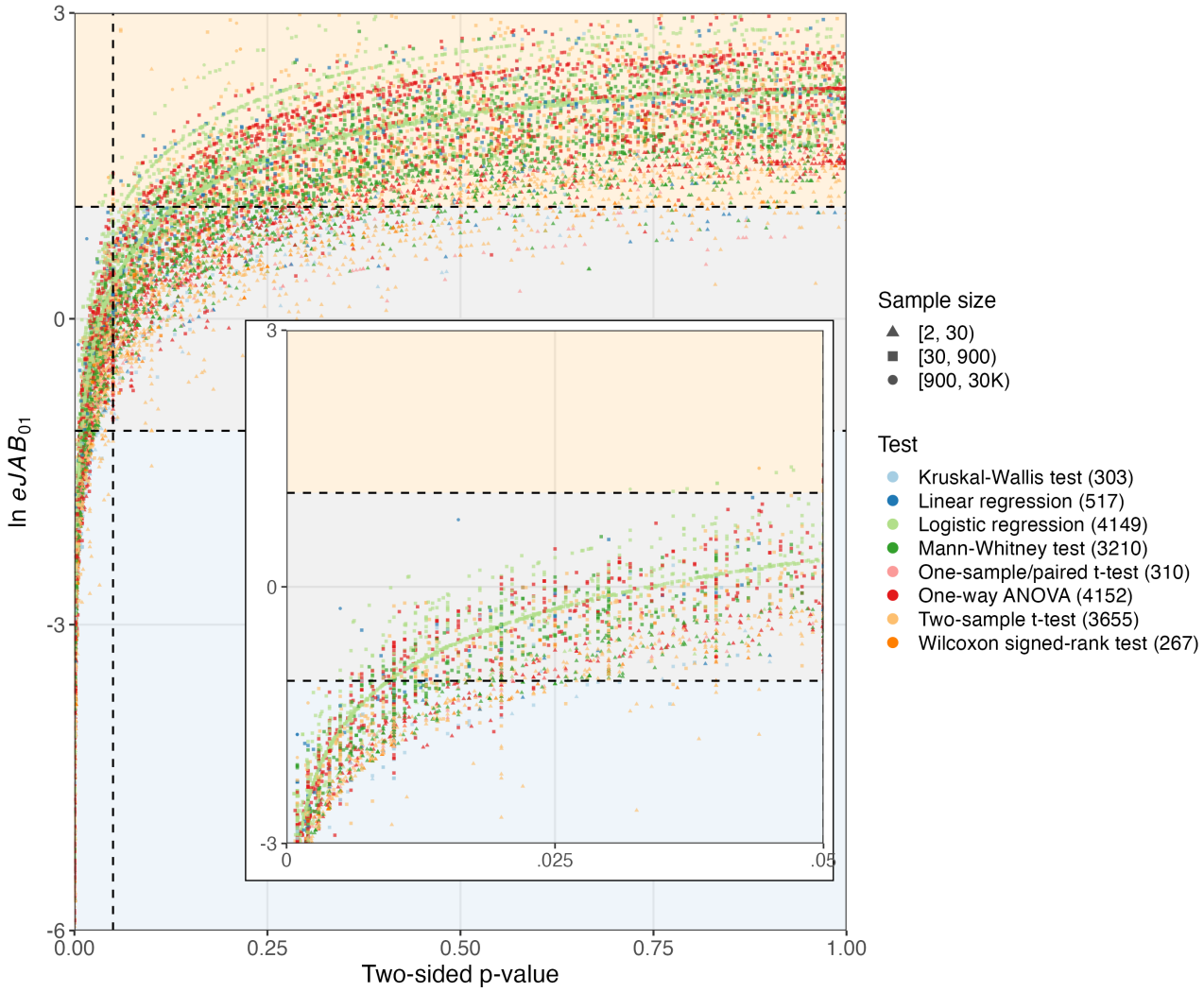


Figure S8: Enlarged version of the scatterplot in Figure S7B. The applicable ln $eJAB$ is plotted against the two-sided p-value, using $p = 2\min\{p_1, 1 - p_1\}$ for one-sided tests. Colors identify statistical tests and shapes identify sample-size groups. Dashed lines mark $p = .05$ and eJAB $= 1/3, 3$. The inset enlarges $p \leq .05$ and $-3 \leq \ln eJAB \leq 3$.